

CSSE 220 – Object-Oriented Software Development
 Rose-Hulman Institute of Technology

Worksheet 07

Name (Print): _____ Section: _____

1. **Instance variable** (Circle the correct answer)

Instance variables belong to a (**class** / **individual object**).

2. **Static variable** (Circle the correct answer)

The keyword **static** indicates that a field belongs to the (**class** / **or individual object**).

3. **Accessing Static vs Instance Fields**

Suppose each **Player** object has an instance field **health**. You have a variable **p** that refers to a **Player** object. How do you access the instance field?

1	
2	

Suppose the class **Player** has a static field **totalPlayers**. How do you access the static field?

1	
2	

4. **THIS Vs Static**

You can use **this** inside a static method (T / F)

5. **Final Keyword** (Circle the correct answer)

The keyword **final** means the variable (**can** / **cannot**) change after the assignment.

6. **Quest Class**. Suppose you have the following class. Circle the correct statements below.

```
public class Quest {
    private String title;
    private final int xpReward;
    private static int totalQuests;
    private static final int MAX_XP = 500;
}
```

- (a) Each **Quest** has its own (**title** / **xpRewards** / **totalQuests** / **MAX_XP**)
- (b) **xpReward** can be assigned: (**once** / **many times**) per object
- (c) **totalQuests** is shared by: (**each object** / **the class**)
- (d) **MAX_XP** is assigned: (**once per object** / **once per class**)
- (e) **MAX_XP** can be modified after assignment: (**yes** / **no**)

7. Overriding

Overriding allows a subclass to change behavior inherited from its superclass. (T / F)

8. Method overriding rules

- The method name must be the same (T / F)
- The return type must be the same (T / F)
- The parameter list must be the same (T / F)
- You may change the number of parameters when overriding (T / F)

9. **Override Example.** Suppose you have a class `Player` with a public field `health`. Write an override of `toString()` that returns "Player's health = 20".

```
1 @Override
2 public String toString() {
3
4     return _____;
5 }
```

10. Encapsulation

Encapsulation means an object _____ its own data by _____
how it is accessed and modified

11. **Getter Method.** Complete the getter method to return the health value:

```
1 public ____ getHealth() {
2
3     return _____;
4 }
```

12. **Setter Method.** Complete the setter method to update the health value:

```
1 public _____ setHealth(int newHealth) {
2
3     _____ = newHealth;
4 }
```

13. Before You Leave

Write one question you still have about today's topic.